

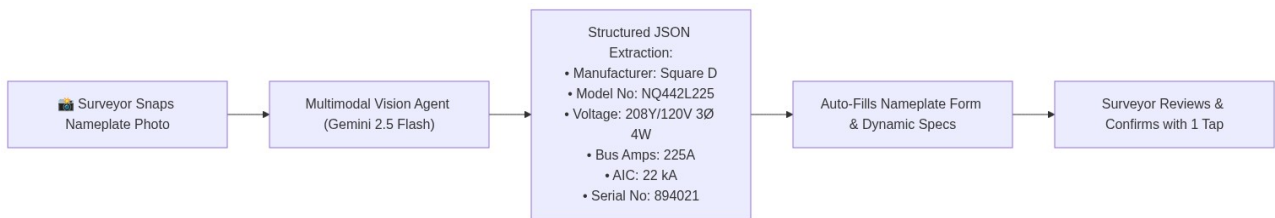
# JEMBY / JAMES: Future Innovations & AI Roadmap

This document outlines the strategic future capabilities, advanced AI integrations, and architectural enhancements planned for the **JEMBY JAMES Electrical Digital Twin Platform**.

## 1. Multimodal AI Nameplate OCR & Field Vision Engine

### 1.1 Overview

Surveyors spend substantial on-site time manually transcribing worn, stamped metal nameplates across hundreds of facility electrical assets. The AI Vision Engine will enable surveyors to snap a single photo of an equipment nameplate, with Gemini Multimodal models automatically parsing, extracting, and populating structured attributes in the Cockpit.



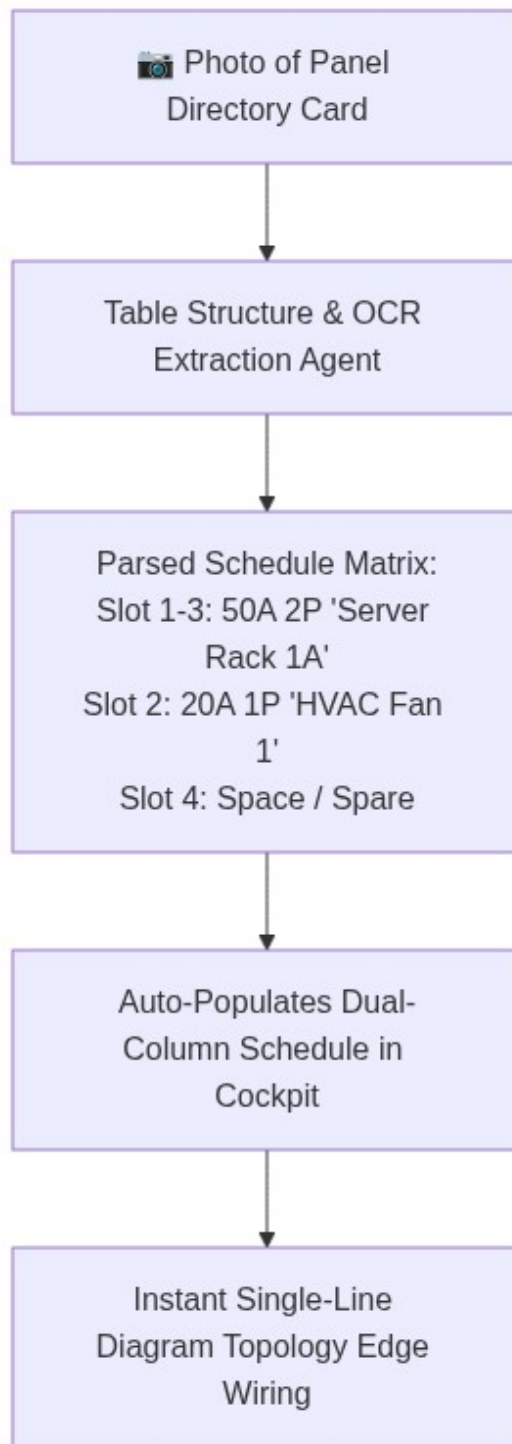
### 1.2 Key Features:

- **Glare & Wear Invariance:** Processes stamped metal plates, embossed lettering, weathered outdoor tags, and dot-peen markings under uneven lighting.
- **Fuzzy Master Catalog Resolution:** Reconciles partially worn model numbers (e.g., NQ44. . . ) against the Master Catalog to complete missing engineering ratings.
- **Contextual Dynamic Attribute Mapping:** Automatically extracts specialized archetype parameters (e.g., %Z = 5.75% for transformers, Prime kW = 750 for generators, or Class R Fuses = 100A for disconnects) directly into vendor\_attrs.
- **Human-in-the-Loop Validation:** Highlights extracted values in the Cockpit UI with visual confidence indicators for instant surveyor verification.

## 2. Panel Directory OCR & Automated Schedule Digitization

### 2.1 Overview

Physical paper panel directories taped inside panelboard doors often contain handwritten or typewritten circuit descriptions, breaker sizes, and load destinations.



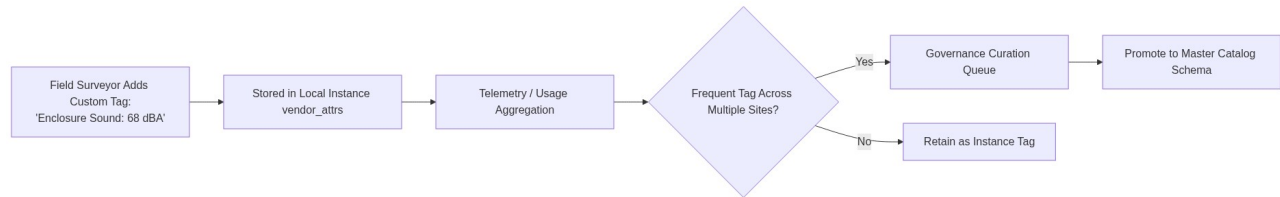
## 2.2 Key Features:

- **Handwritten & Typewritten Text Recognition:** Reads handwritten circuit descriptions and pencil marks.
- **Slot Alignment & Multi-Pole Detection:** Distinguishes 1-pole, 2-pole, and 3-pole tied breakers and assigns them to odd/even left/right slot positions.
- **Automatic Downstream Load Mapping:** Parses circuit names (e.g., AHU-3, Lighting ER-101, Chiller Pump 2) and automatically matches them to existing equipment in the stack.

### 3. Master Catalog Governance & Field Back-Propagation

#### 3.1 Overview

When field surveyors discover and record novel equipment attributes using the ■ **Add Field Attribute** tool, these tags initially attach strictly to that individual asset instance. This preserves catalog hygiene while offering infinite field flexibility.



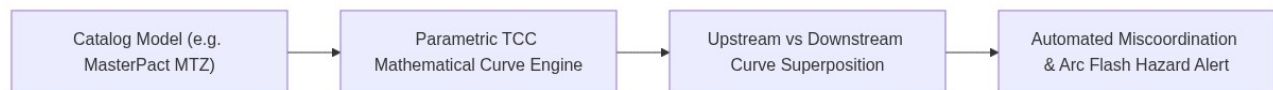
#### 3.2 Key Features:

- **Per-Instance Isolation:** Ensures field creativity does not pollute global schemas or cause breaking database alterations.
  - **Telemetry & Tag Frequency Heatmaps:** Identifies recurring custom parameters across client portfolios.
  - **One-Click Promotion Workflow:** Catalog administrators can formally promote validated attributes into the global Master Catalog.
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### 4. Time-Current Characteristic (TCC) & Selective Coordination

#### 4.1 Overview

Integration between the Master Catalog breaker models and standardized Time-Current Curves (TCC) to automate electrical protection engineering.



#### 4.2 Key Features:

- **Manufacturer Trip Curve Library:** Stores Long-Time (L), Short-Time (S), Instantaneous (I), and Ground-Fault (G) settings for Square D, Eaton, Siemens, and ABB trip units.
  - **Automated Overlap Detection:** Warns engineers if a downstream branch breaker will trip upstream main protection.
  - **Arc Flash Boundary Estimation:** Uses IEEE 1584 calculations based on breaker clearing times and short-circuit current.
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### 5. Infrared (IR) Thermography & Field Sensor Integration

#### 5.1 Overview

Direct attachment of thermal imaging camera data (FLIR, Seek Thermal) to circuit breaker slots, transformer coils, and cable lugs.

## 5.2 Key Features:

- **Radiometric Photo Embedding:** Stores thermal images with temperature metadata per slot.
  - **Delta-T Thermal Anomaly Flagging:** Automatically calculates  $\Delta T$  against adjacent phases and highlights overheating lugs in the SLD and Panel Schedule.
  - **Preventive Maintenance Ticketing:** Generates immediate work orders for loose connections or phase load imbalances.
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## 6. Offline-First Progressive Web App (PWA) Field Mode

### 6.1 Overview

Full offline operation for industrial basements, electrical vaults, mines, and remote substations with zero internet connectivity.

### 6.2 Key Features:

- **Local SQLite / WASM Storage:** Full client-side database running in the browser engine via WebAssembly.
- **Background Delta Sync:** Automatically reconciles and merges offline field surveys back into the central cloud repository when connectivity is restored.
- **Compressed Image Queuing:** Efficiently manages high-resolution nameplate and inspection photos offline before uploading.